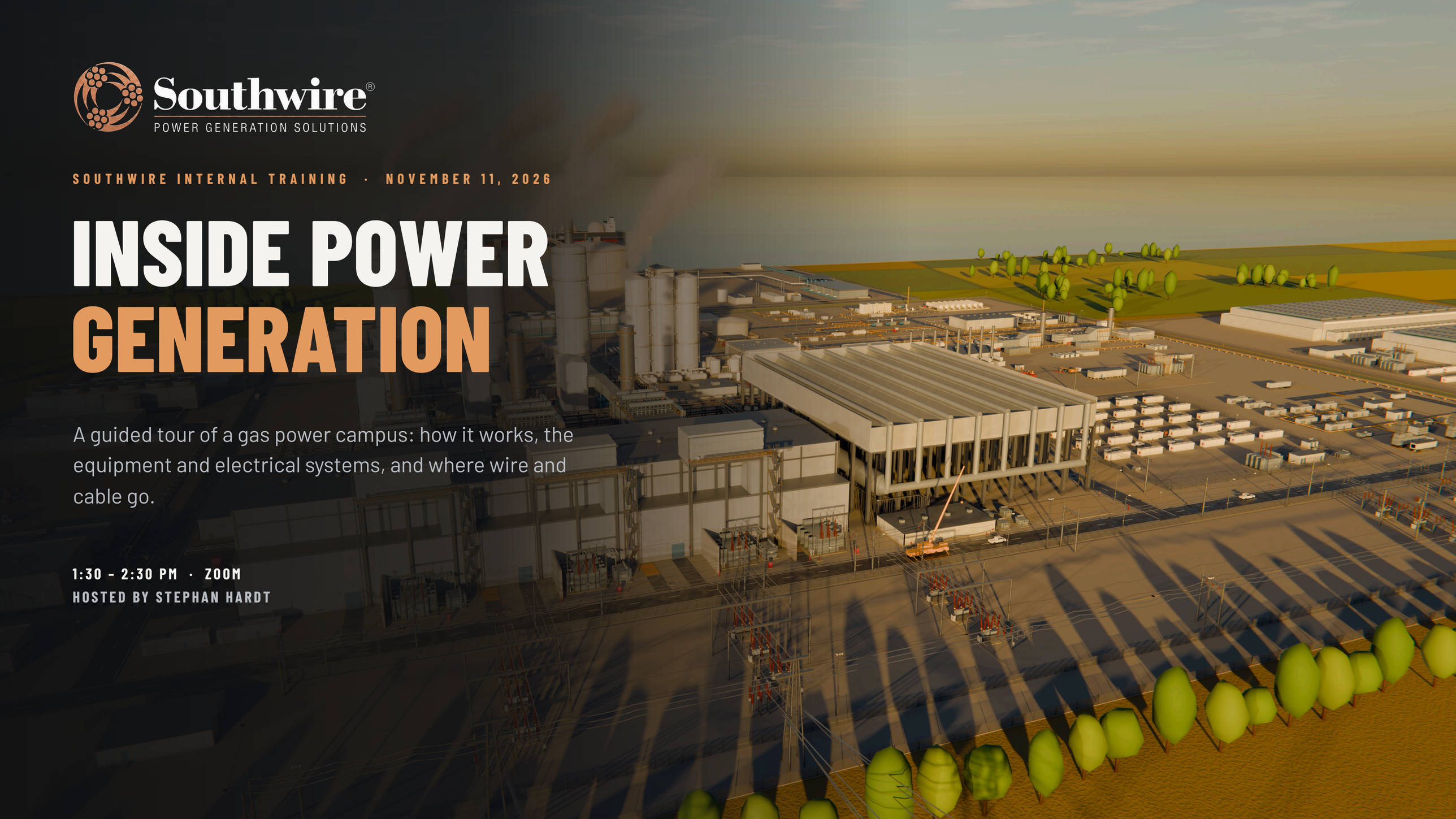
Southwire®
POWER GENERATION SOLUTIONS

SOUTHWIRE INTERNAL TRAINING · NOVEMBER 11, 2026

INSIDE POWER GENERATION

A guided tour of a gas power campus: how it works, the equipment and electrical systems, and where wire and cable go.

1:30 - 2:30 PM · ZOOM
HOSTED BY STEPHAN HARDT



TODAY · 60 MINUTES

THE HOUR

1:30	**WELCOME** why this matters	3 MIN
1:33	**MARKET AND FUNDAMENTALS** the gas build and the cable prize; the black box, owners, the POI corridor	18 MIN
1:51	**THE 3D TOUR** seven stops: what it does, its electrical system, its cable	22 MIN
2:13	**WIRE AND CABLE TAKEAWAYS** the whole plant in one number, products, pain points	9 MIN
2:22	**Q&A** questions from the chat	8 MIN

No engineering background needed. At every stop you get three things: **what it does**, its **electrical system**, and **the cable**.

Questions in the chat; we answer them at the end.

stephan.hardt@southwire.com · INSIDE POWER GENERATION 2/26

A BOOM-SCALE GAS BUILD, AND CABLE IS ON THE CRITICAL PATH

THE MARKET

378 GW

US gas power in development; 52 GW already under construction

THE CABLE

~$1.2B

core wire & cable, 2026–2035, at $8–13M per GW

THE PIPELINE

386

gas projects we track in 40 states, every owner type

THE TRACTION

$50M+

booked in Q3; $30M in September alone

TODAY'S GOAL: KNOW THE PLANT, SO YOU CAN FOLLOW THE CABLE THROUGH IT

Global Energy Monitor, Aug 2026; Southwire PowerGen planning model and internal figures, 2026.

stephan.hardt@southwire.com

MARKET CONTEXT

LOAD GROWTH IS PULLING GAS POWER FORWARD

166 GW

US peak load growth forecast by 2030

about 90 GW of it from data centres

378 GW

US gas power in development, mid-2026

189 GW tied to data centres

52 GW

under construction, up 76% in H1 2026

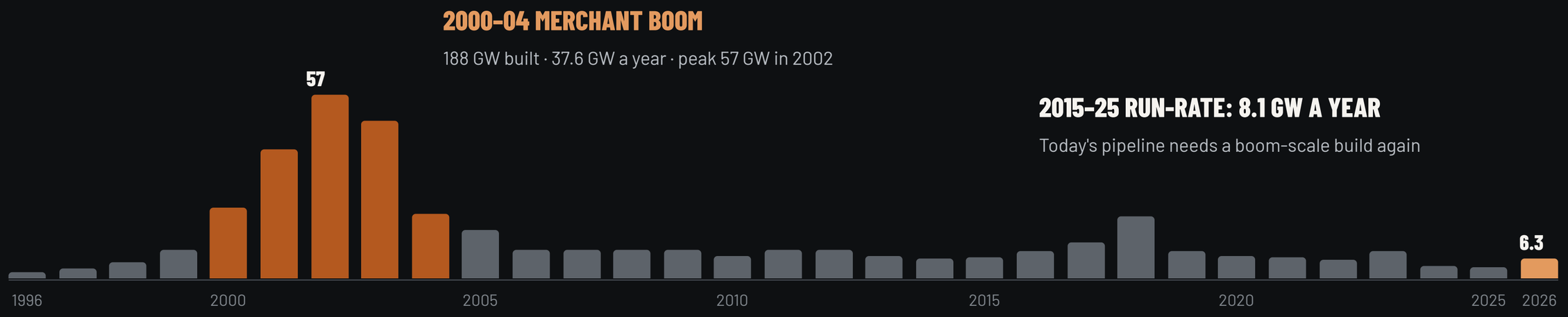
the largest gas build in the world

Up to 7 yrs

quoted lead time for some large gas turbines

procurement decisions are being made now

US GAS CAPACITY ADDED PER YEAR, GW · PROVISIONAL



Grid Strategies, 2025; Global Energy Monitor, Aug 2026; S&P Global, May 2025. History: EIA and Southwire provisional series, to be verified.

stephan.hardt@southwire.com

OEM ORDER BOOKS

TURBINE BACKLOGS ARE FULL



The three large-frame turbine OEMs together hold **about 220 GW** of gas turbines in backlog and slot reservations. Modular and packaged gas power is growing beside them.

MODULAR

+72%

power-generation sales at the largest engine-generator maker, Q2 2026

BEHIND THE METER

2.3 GW

modular gas fleet for Texas AI data centres, plus 1 GW for another campus

PACKAGED

500 MW

standard modular gas-turbine plant, offered as a product

OEM releases, Q2–Q3 2026 (firm backlog plus slot reservations; gas-turbine backlog; large-frame backlog); Utility Dive, Aug 2026; POWER, Oct 2025 and Feb 2026.

stephan.hardt@southwire.com

SIZING THE OPPORTUNITY · CONSERVATIVE PLANNING CASE

~$1.2B OF CORE GAS CABLE, 2026–2035

CAPACITY

93.7 GW

modeled additions through 2036, from the 378 GW pipeline: 90% of the 52 GW building, 13.5% of the rest

PLANT INVESTMENT

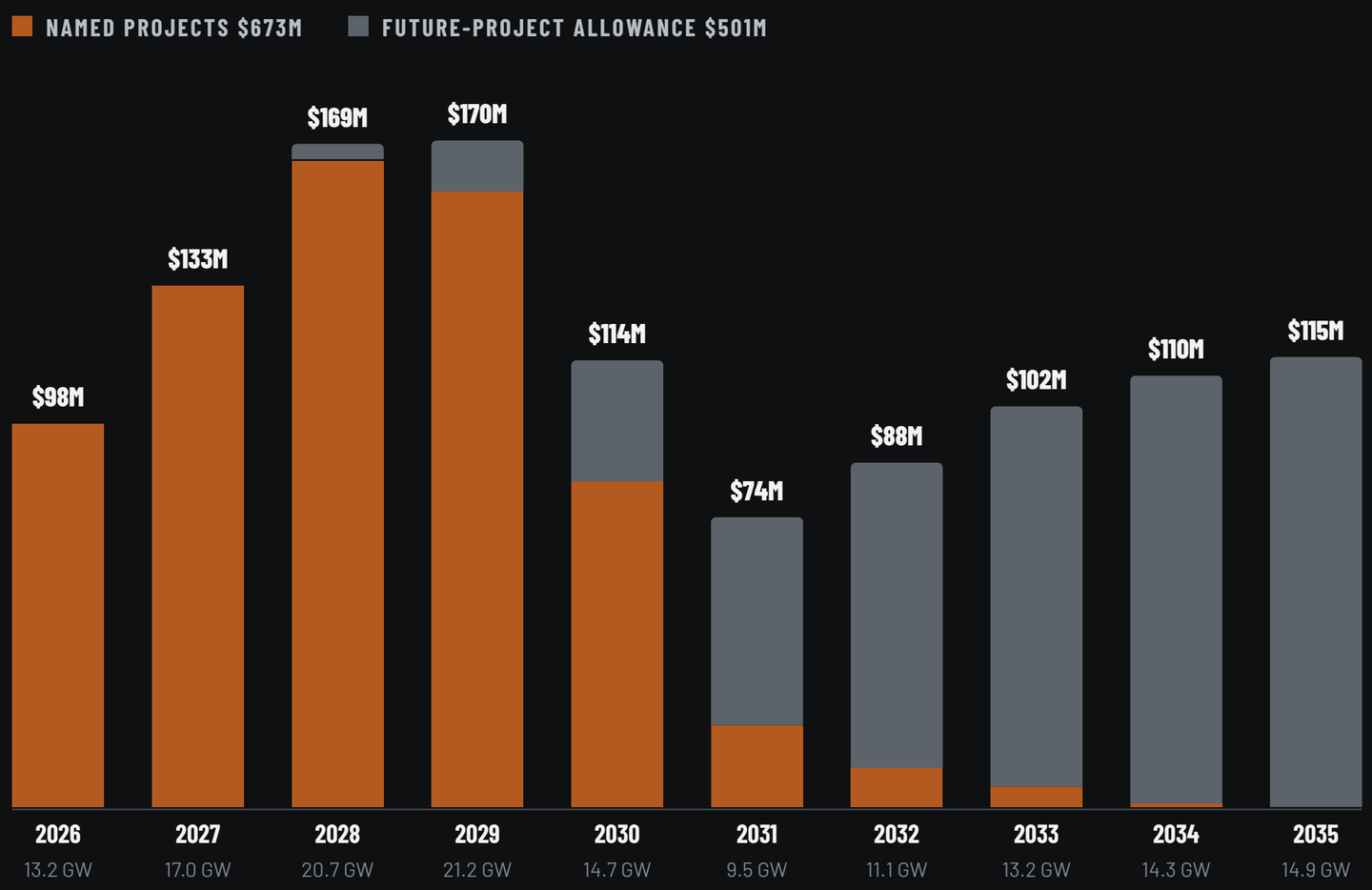
~$180B

93.7 GW × $1,923/kW blended (combined cycle, peakers, engines)

WIRE & CABLE

$0.75–1.2B

$8–13M per GW: under 1% of plant capex, on the critical path to first power



Core cable by purchase-order year; below each year, modeled PO-year GW (not installed GW). GW × $/kW × addressable share × win probability. OEM internal wiring and HV excluded.

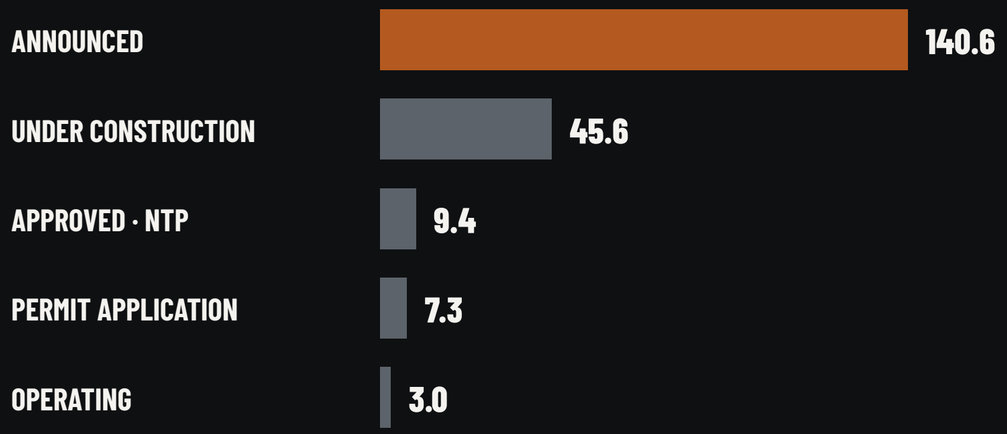
Global Energy Monitor, Aug 2026; completion rates, blended cost and cable rates are Southwire planning assumptions.

stephan.hardt@southwire.com · INSIDE POWER GENERATION 6/26

THE PROJECT REGISTER

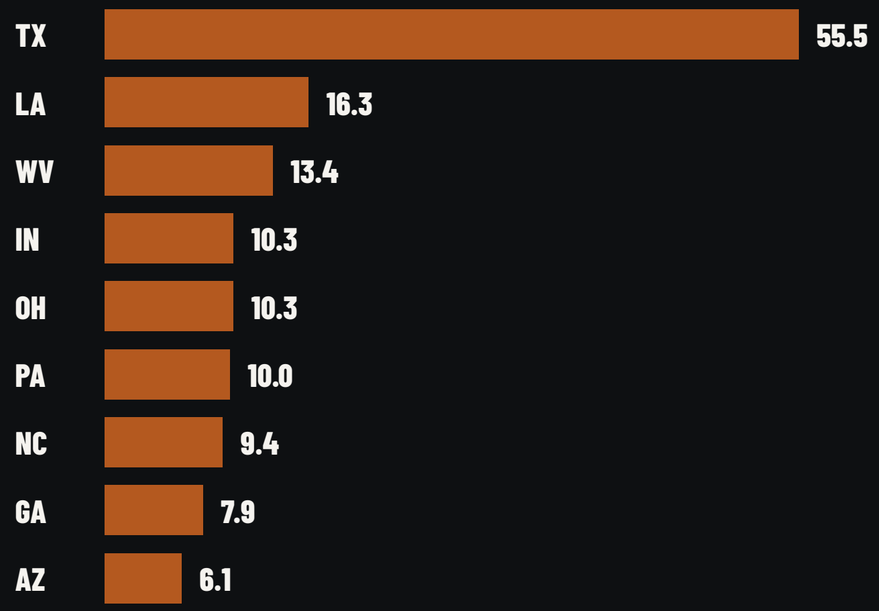
205.8 GW ELIGIBLE, MOST OF IT STILL EARLY

BY STATUS · GROSS GW, COD 2026-2035



68% **of eligible capacity is only announced.** Being on the register does not mean a cable package is open; it means a conversation to start.

TOP STATES · GROSS GW, 386 PROJECTS



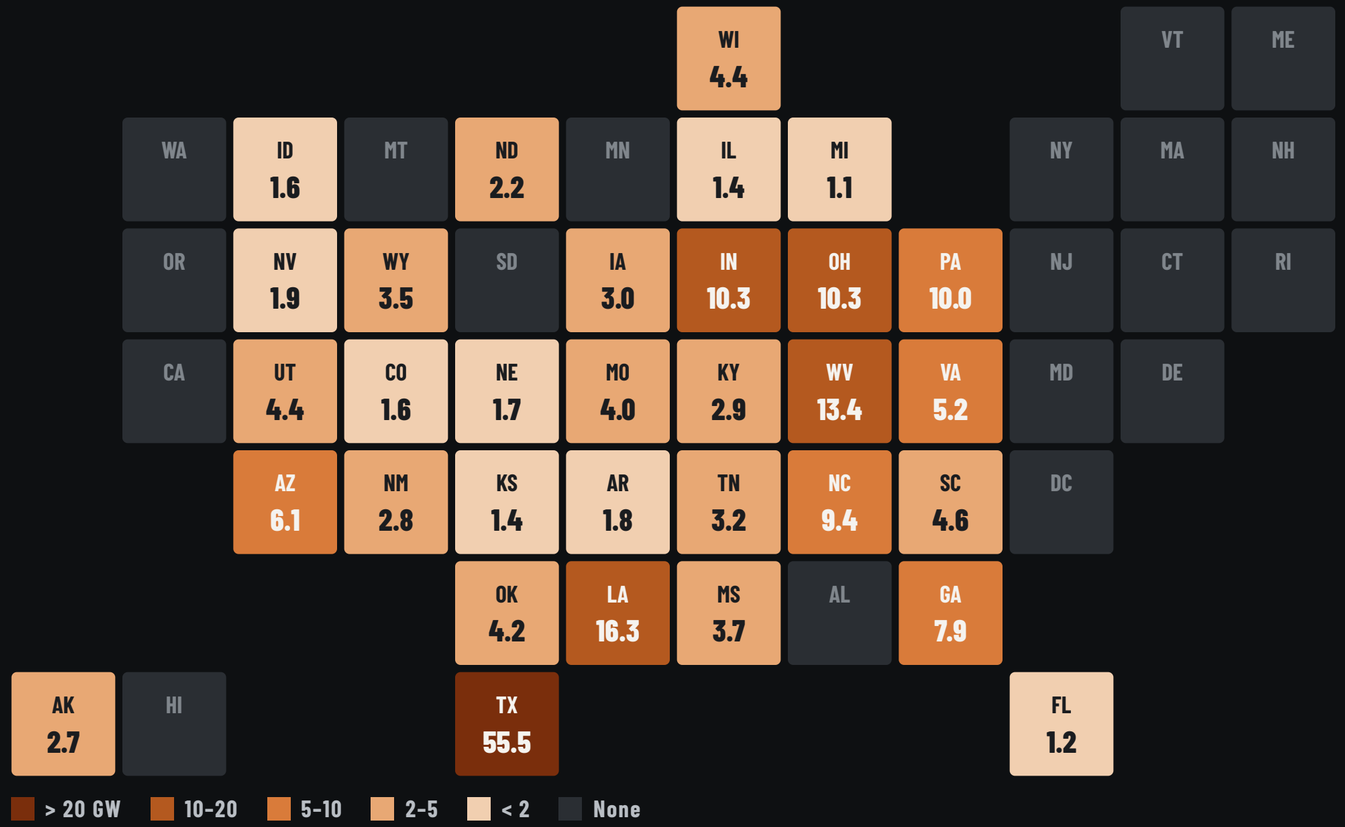
Greenfield 137 GW, brownfield 69 GW. Utilities reuse old sites (coal-site repowers, expansions); independent and behind-the-meter developers mostly build new, often on data-centre campuses.

Southwire PowerGen register, Sep 11, 2026: 386 records across 305 site groups; gross GW, unweighted.

PROJECT CONCENTRATION

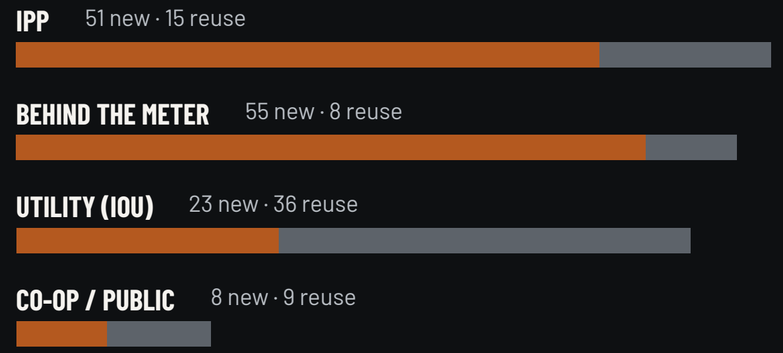
WHERE THE BUILD IS, WHO OWNS IT, WHAT KIND OF SITE

GROSS GW BY STATE · 386 PROJECTS, COD 2026-2035



OWNER × SITE · GROSS GW

GREENFIELD 137 · BROWNFIELD 69



UTILITIES REUSE OLD SITES; IPPS AND BTM BUILD NEW

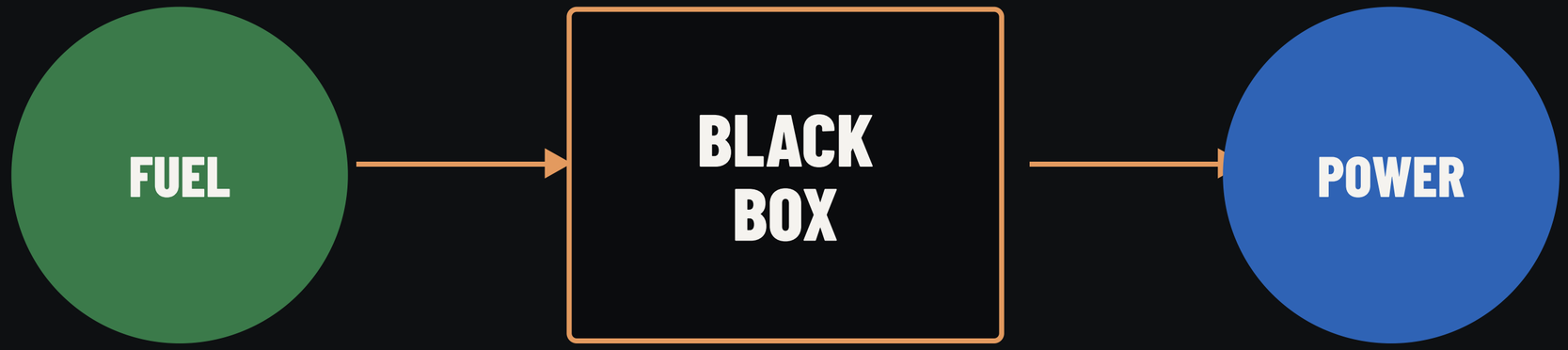
Greenfield, grid-connected	77.0
Data-centre / BTM campus	54.5
Coal-site repower	41.3
Expansion at an operating plant	21.4
Industrial captive and site reuse	11.6

Southwire PowerGen register, Sep 11, 2026. Gross GW, unweighted. Site classes provisional (Rev01).

stephan.hardt@southwire.com · INSIDE POWER GENERATION 8/26

EVERY THERMAL PROJECT IS THIS

Input → conversion → output. The complexity is what's inside the box, and what's around it.



INPUT
fuel / energy source

Thermal conversion

OUTPUT
dispatchable power

Gas turbine, combined cycle or engines: **the same physics on every project.** Today we open the box and follow the power out of it.

OPEN THE BOX: THE PHYSICAL ANATOMY

From fuel infrastructure to dispatchable electricity and end-use demand.

0

FUEL INFRASTRUCTURE

- Gas production, pipelines, compression
- Storage and LNG logistics
- Plant-gate fuel delivery

1

FUEL INPUT

- Natural gas, coal, oil / diesel
- Hydrogen blends
- Biomass where it applies

2

THERMAL CONVERSION

- Frame and aero gas turbines, engines (RICE)
- Combined cycle: GT + HRSG + ST
- Modular and H2-ready packages

3

ELECTRICITY OUTPUT

- Dispatchable AC power
- Stable, reliable load
- Baseload, mid-merit or peaking

4

DELIVERY & FLEXIBILITY

- Transmission and distribution
- MV / LV delivery, behind-the-meter on site
- Storage and flexibility support

5

END USE / OFF-TAKE

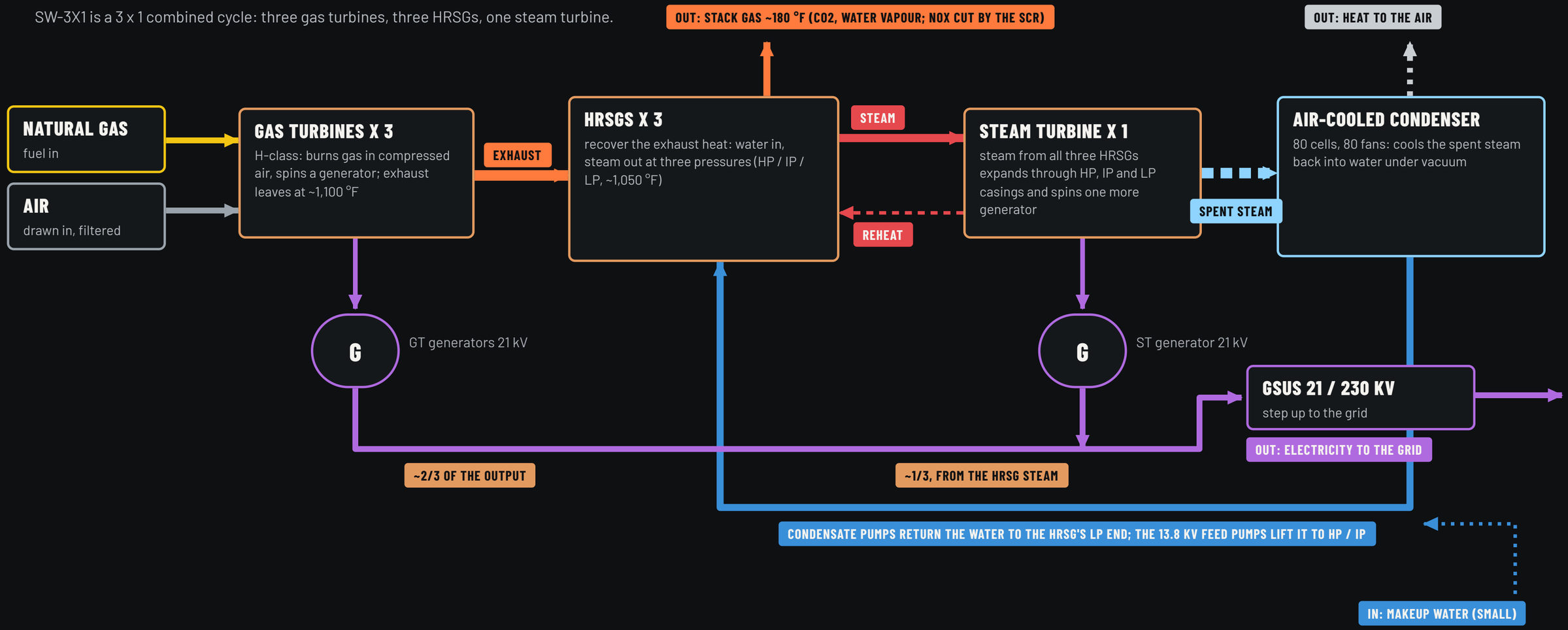
- Utilities, IPPs, industrial loads
- Data centres and other off-takers
- Critical infrastructure

Same physics on every project: gas turbine, combined cycle or engines. **What changes is who owns it.**

stephan.hardt@southwire.com

WHAT GOES IN, WHAT COMES OUT

SW-3X1 is a 3 x 1 combined cycle: three gas turbines, three HRSGs, one steam turbine.



IS IT STILL STEAM?

Leaving the turbine it is spent steam: ~100-150 °F, below atmospheric pressure, part water already. The air-cooled condenser turns it back into water.

A CLOSED LOOP

The same water goes round and round: pump, HRSG, turbine, condenser. Only a little makeup water replaces blowdown losses.

WHERE THE POWER COMES FROM

Fuel is burned only in the gas turbines (and duct burners). The steam turbine adds about half again on top, from heat that would otherwise go up the stack.

THE CABLE ON THE LOOP

21 kV isolated-phase bus to the GSUs; MV-105 to the feed pumps; VFD cable to 80 condenser fans; instruments on every pump, valve and level.

stephan.hardt@southwire.com · INSIDE POWER GENERATION 11/26

FUNDAMENTALS · THE BASICS

HOW AN HRSG WORKS

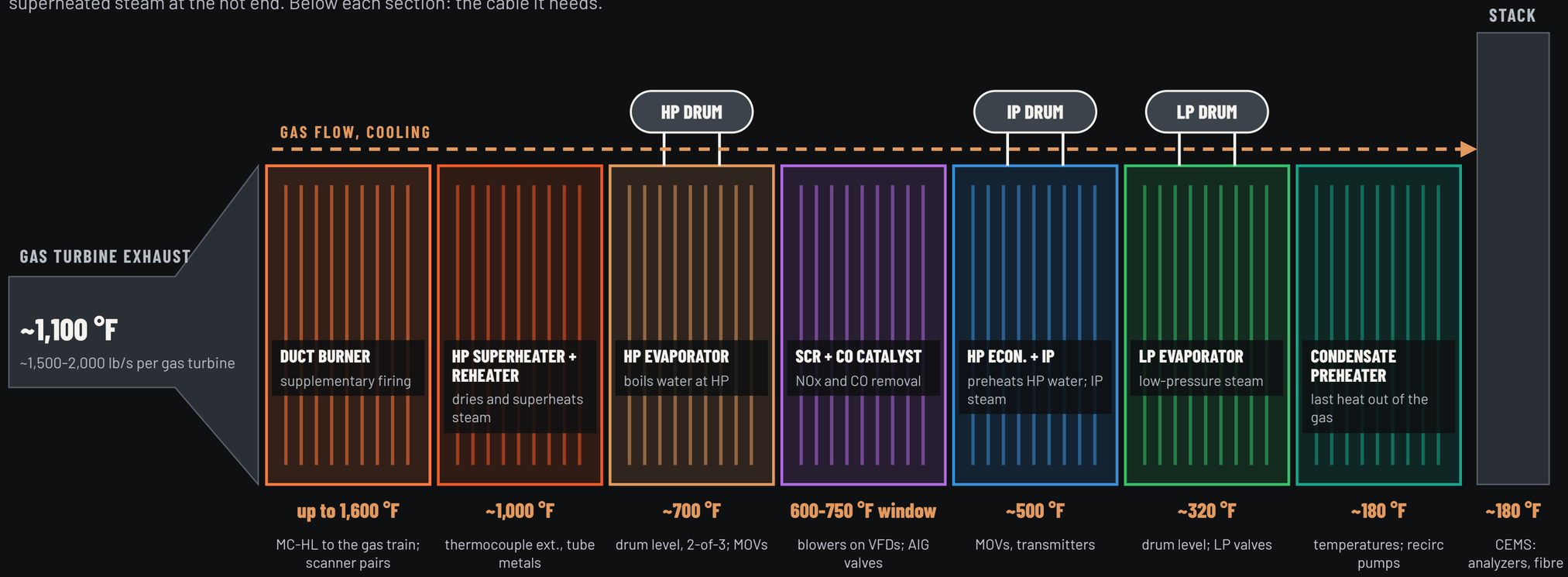
The exhaust crosses rows of finned tubes, hottest sections first; water enters at the cold end and leaves as superheated steam at the hot end. Below each section: the cable it needs.

WHAT IT IS

- **Without it:** ~60 % of the fuel's energy goes up the stack as ~1,100 °F exhaust (~40 % efficient).
- **With it:** a boiler heated by that exhaust; its steam drives the steam turbine.
- **The payoff:** ~1/3 more power from the same fuel (~60 %+ efficient).

FUN FACTS

- Fuel, CO2 and cost per MWh drop by about a third.
- The NOx catalyst (SCR) lives **inside** it, where the gas is just the right temperature.
- Duct burners can fire extra gas for peak demand.
- No HRSG: a gas-turbine plant. With one: combined cycle.



Steam to the steam turbine, typical: HP ~1,800-2,400 psig, ~1,050 °F · reheat / IP ~400-600 psig, ~1,050 °F · LP ~50-80 psig

~1/3
of a combined-cycle plant's output comes from HRSG steam, with no extra fuel (duct firing aside)

~60 %+
combined-cycle efficiency, against ~40 % for the gas turbine alone

3 x 3
three pressures, three drums; each drum level measured three ways, typically voted 2-out-of-3

~1 hour
a hot start: about an hour: every start cycles the tubes, which is why tube-metal thermocouples matter

Typical three-pressure reheat HRSG with SCR behind an H-class gas turbine; section order, temperatures and pressures vary by design.

stephan.hardt@southwire.com · INSIDE POWER GENERATION 12/26

SAME PHYSICS, DIFFERENT OWNERS

The owner decides how the project is structured, and how it gets bought.

THE STRUCTURAL EQUATION

THERMAL IPP = **PROJECTCO** (SPV / LLC) + **ASSET** (conversion plant) + **CONTRACT** (PPA / off-take) + **CUSTOMER** (off-taker / load) + **CAPITAL** (PE / tax equity) + **MARKET** (ERCOT, PJM, MISO)

POWER OWNER ARCHETYPES

IOUS / CO-OPS — REGULATED

- Utility-led, rate-based ownership
- Own and operate generation
- Buy like utilities: standards, cycles

HYBRID IPPS — MIXED MODEL

- Utility-affiliated IPP subsidiaries
- Bridge regulated and competitive
- Two buying behaviours

PURE IPPS — COMPETITIVE

- Privately owned, PE-backed projects
- A direct fit with the project equation
- Speed and cost drive decisions

PRIMARY FOCUS FOR COMPETITIVE THERMAL POWER

WHERE SOUTHWIRE WINS: THE POI CORRIDOR

The cable story lives between electricity output and the grid: plant electrical, gen-tie and utility delivery.

0 FUEL INFRA

1 FUEL INPUT

2 THERMAL CONVERSION

Upstream: not our fight

→

THE SOUTHWIRE CORRIDOR

3 ELECTRICITY OUTPUT

- Generator leads
- Power collection and busbar
- Grounding systems

4A PLANT ELECTRICAL & INTERCONNECTION

- GSU transformer, switchyard
- MV / HV cable and tray
- E-houses, controls, BESS tie-ins

4B UTILITY / GRID DELIVERY

- Gen-tie to the POI
- Transmission and distribution cable
- Substation solutions

HV UNDERGROUND: GEN-TIE AND PLANT-TO-POI | **POI**

Plant / EPC scope

Utility / transmission-owner scope

Every MW that leaves the black box crosses this corridor. Cable intensity concentrates at 3 → 4A → 4B, the same span where the POI boundary blurs scope, and where we help EPCs and IPPs.



THE CAMPUS · FROM THE SEA TO THE GRID

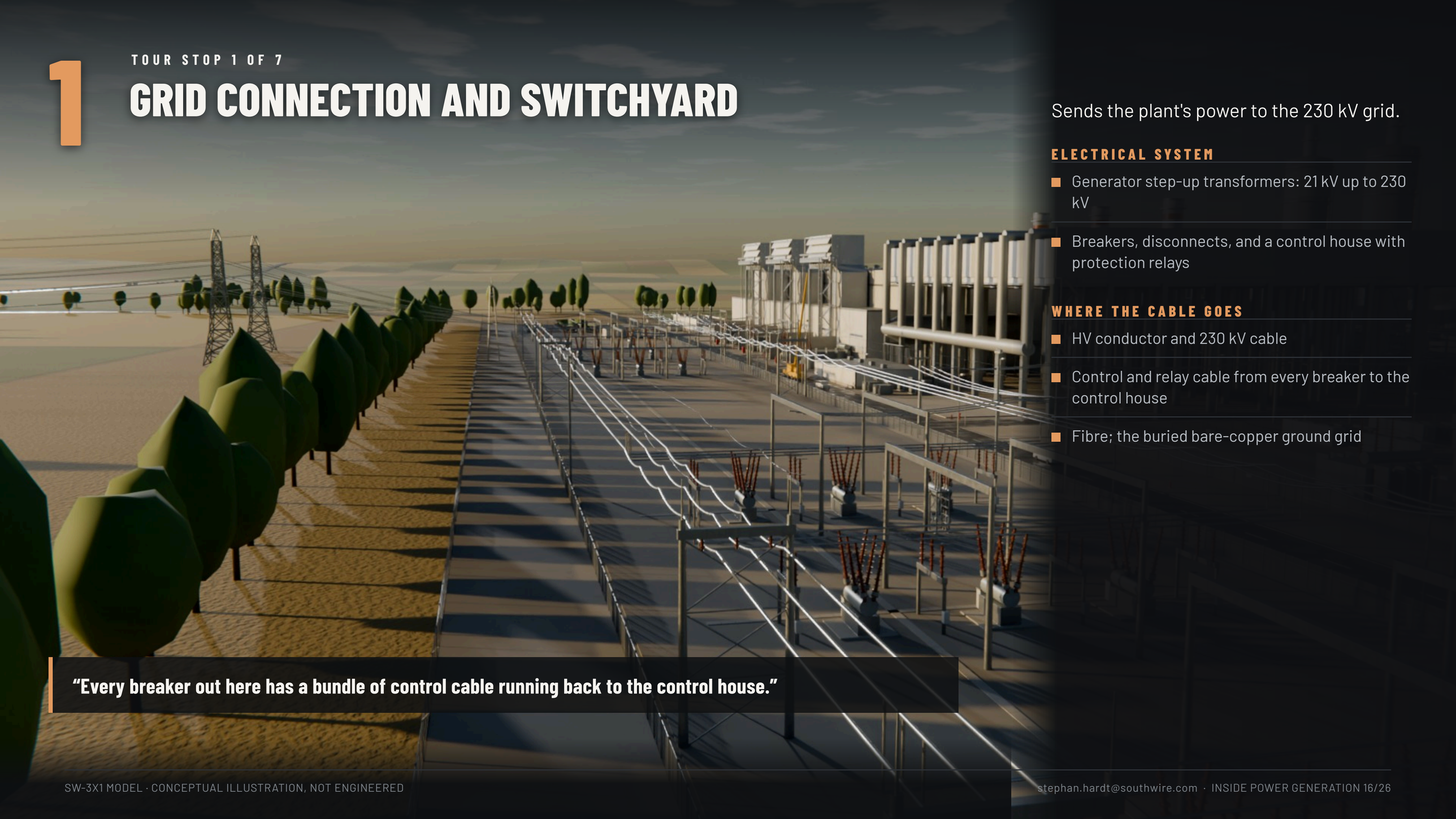
ONE SITE, SEVEN STOPS

1. GRID CONNECTION AND SWITCHYARD
2. TURBINE HALL
3. HRSG AND ITS E-HOUSE
4. COOLING AND WATER
5. MODULAR GENERATION
6. BEHIND THE METER: THE DATA CENTRE
7. FUEL: GAS YARD AND LNG

LNG arrives by ship at the marine terminal; the power leaves on 230 kV lines. Everything in between runs on cable.

TOUR STOP 1 OF 7

1 GRID CONNECTION AND SWITCHYARD



Sends the plant's power to the 230 kV grid.

ELECTRICAL SYSTEM

- Generator step-up transformers: 21 kV up to 230 kV
- Breakers, disconnects, and a control house with protection relays

WHERE THE CABLE GOES

- HV conductor and 230 kV cable
- Control and relay cable from every breaker to the control house
- Fibre; the buried bare-copper ground grid

“Every breaker out here has a bundle of control cable running back to the control house.”



TOUR STOP 2 OF 7

2 TURBINE HALL

Three H-class gas turbines and one steam turbine turn the generators.

ELECTRICAL SYSTEM

- Isolated-phase bus from the generators to the step-up transformers
- Unit auxiliary transformers feed the 13.8 kV switchgear
- Static starters spin the gas turbines up

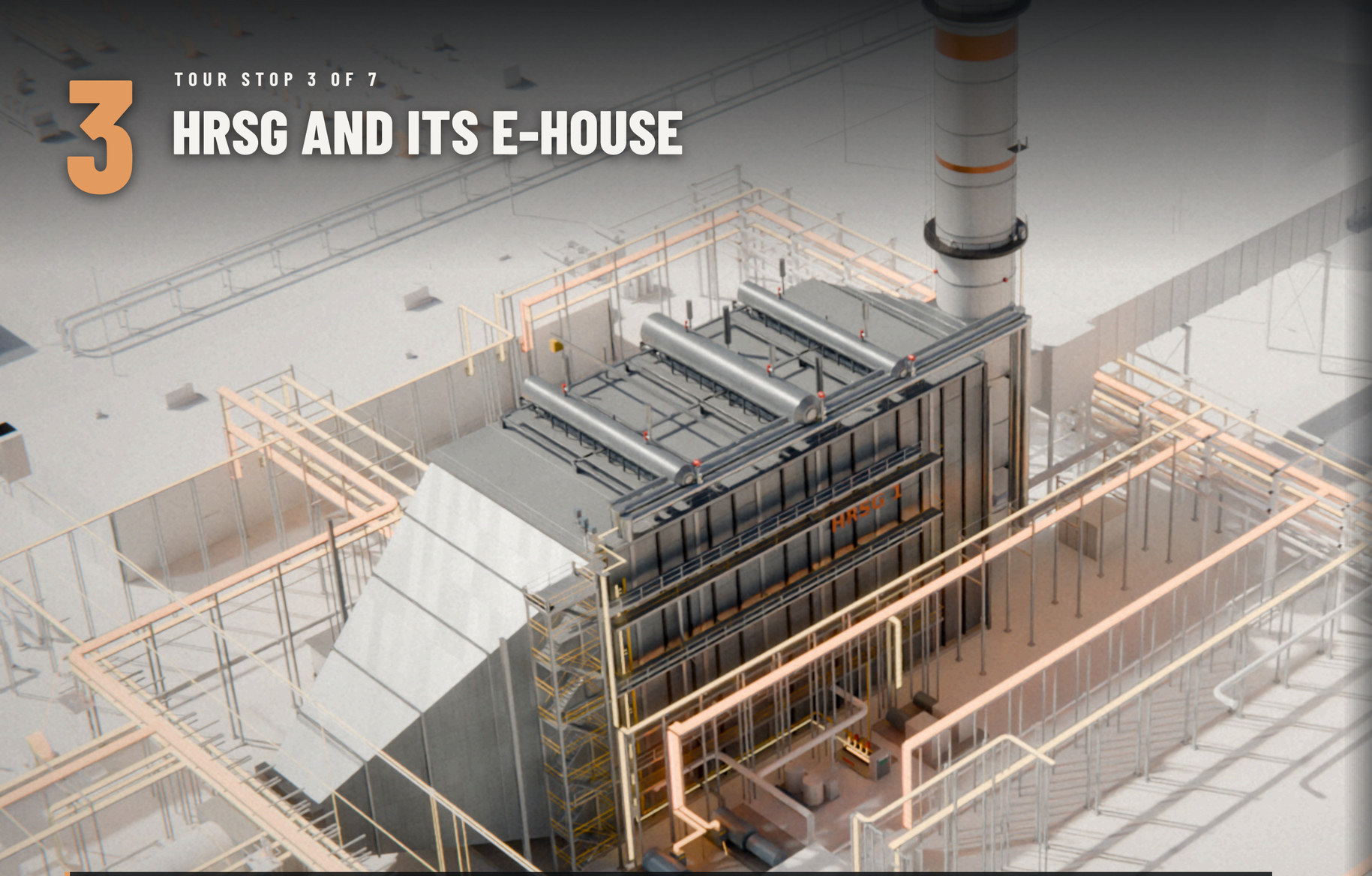
WHERE THE CABLE GOES

- MV-105 to big motors and transformers
- TC-ER power and control to the lube-oil and hydraulic skids
- Thousands of instrument pairs: vibration, temperatures

“The machines make the power; the cable keeps them running and tells us how they feel.”

TOUR STOP 3 OF 7

3 HRSG AND ITS E-HOUSE



Turns turbine exhaust heat into steam; the SCR inside cuts NOx.

ELECTRICAL SYSTEM

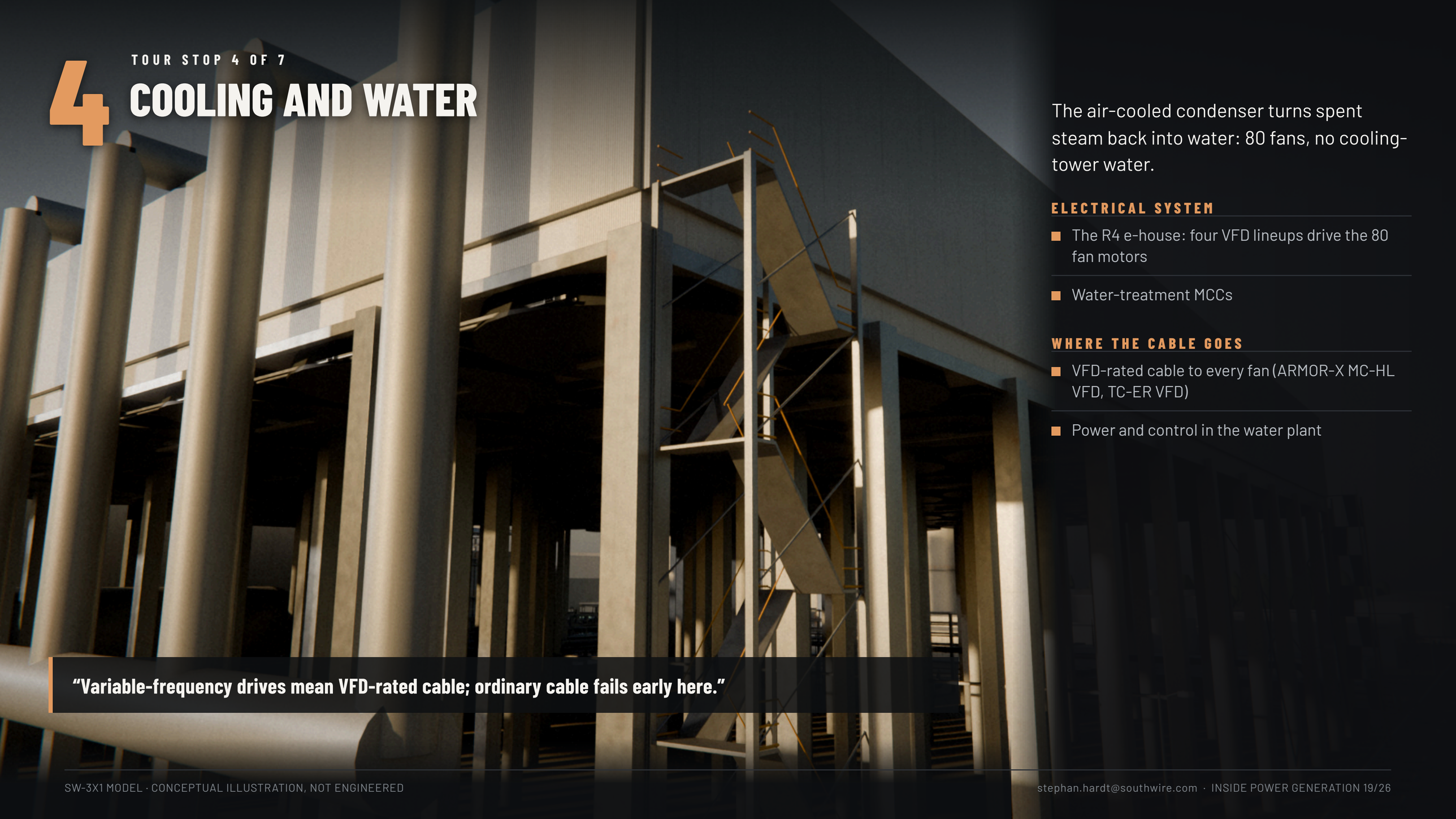
- Two 13.8 kV feed pumps
- A transformer feeds the e-house: 480 V switchgear, MCCs, drives, DCS
- CEMS shelter at the stack

WHERE THE CABLE GOES

- One HRSG train: 707 cables, about 157,000 ft
- About three quarters are instrument and thermocouple pairs
- ARMOR-X MC-HL at the duct-burner gas skid

“One boiler train is about 30 miles of cable, and most of it is small instrument cable.”

SW-3X1 MODEL · CONCEPTUAL ILLUSTRATION, NOT ENGINEERED



TOUR STOP 4 OF 7

4 COOLING AND WATER

The air-cooled condenser turns spent steam back into water: 80 fans, no cooling-tower water.

ELECTRICAL SYSTEM

- The R4 e-house: four VFD lineups drive the 80 fan motors
- Water-treatment MCCs

WHERE THE CABLE GOES

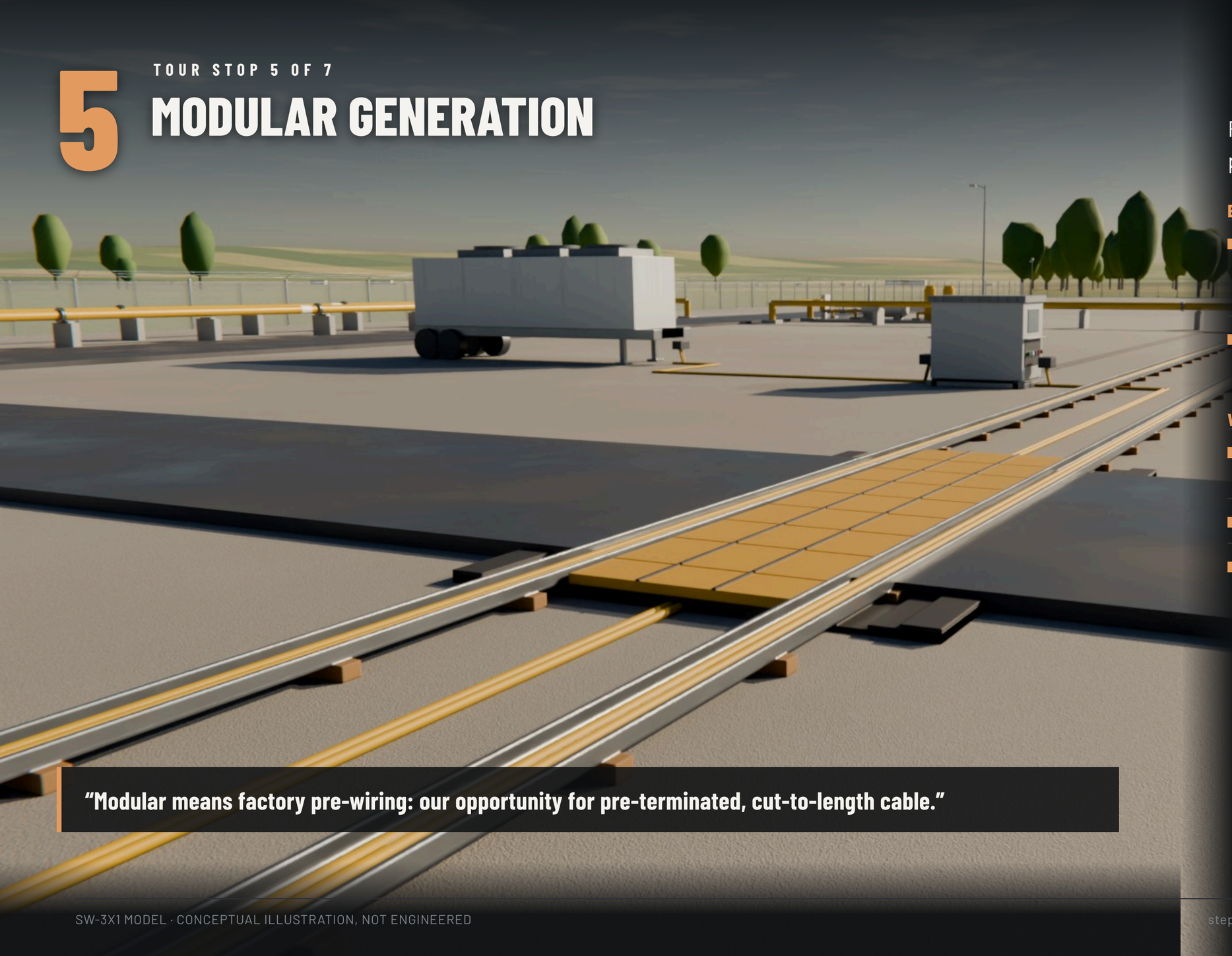
- VFD-rated cable to every fan (ARMOR-X MC-HL VFD, TC-ER VFD)
- Power and control in the water plant

“Variable-frequency drives mean VFD-rated cable; ordinary cable fails early here.”

stephan.hardt@southwire.com · INSIDE POWER GENERATION 19/26

TOUR STOP 5 OF 7

5 MODULAR GENERATION



Fast-start, flexible power beside the big plant.

ELECTRICAL SYSTEM

- Gas engines in a hall, aeroderivative and industrial gas-turbine packages, containers, trailers, fuel cells, microturbines
- Each unit on the 13.8 kV collector; inverters for the battery and fuel cells

WHERE THE CABLE GOES

- MV collection cable; DC cable at the battery and fuel cells
- Factory pre-wired skids and plug-in modules
- Pre-terminated and cut-to-length cable

“Modular means factory pre-wiring: our opportunity for pre-terminated, cut-to-length cable.”

SW-3X1 MODEL · CONCEPTUAL ILLUSTRATION, NOT ENGINEERED

TOUR STOP 6 OF 7

6 BEHIND THE METER: THE DATA CENTRE



A data-centre campus powered directly by the plant's modular units, with the grid as backup.

ELECTRICAL SYSTEM

- 13.8 to 34.5 kV step-up transformers and double-bus switchgear
- A 40 MW battery absorbs load steps
- Two data halls fed 2N through pad-mount transformers and UPS

WHERE THE CABLE GOES

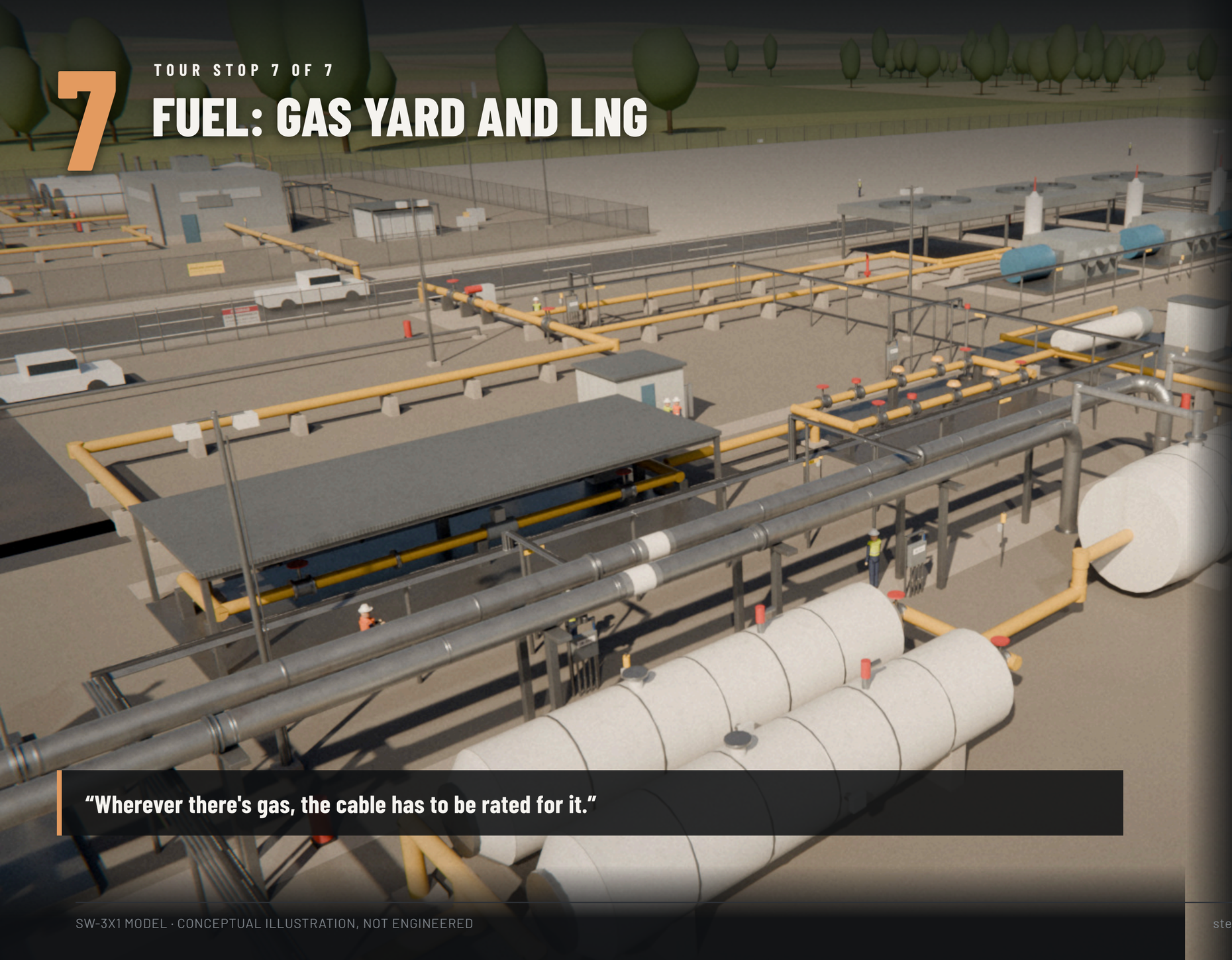
- 35 kV MV cable
- Large LV feeders to the UPS skids
- Fibre everywhere

> **“Data centres are why gas is growing again, and they are cable-dense on both the A and the B side.”**

stephan.hardt@southwire.com · INSIDE POWER GENERATION 21/26

TOUR STOP 7 OF 7

7 FUEL: GAS YARD AND LNG



Brings the fuel in and conditions it; an LNG satellite holds about three hours of backup fuel.

ELECTRICAL SYSTEM

- Metering and regulating station; the LNG e-house (13.8 kV / 480 V)
- Pumps, vaporizers, boil-off compressor

WHERE THE CABLE GOES

- ARMOR-X MC-HL for Class I Div 1 and 2
- TC-ER control, PLTC instrumentation and fire alarm on one strut rack

> **“Wherever there's gas, the cable has to be rated for it.”**

stephan.hardt@southwire.com · INSIDE POWER GENERATION 22/26

THE WHOLE PLANT IN ONE NUMBER

6.4M

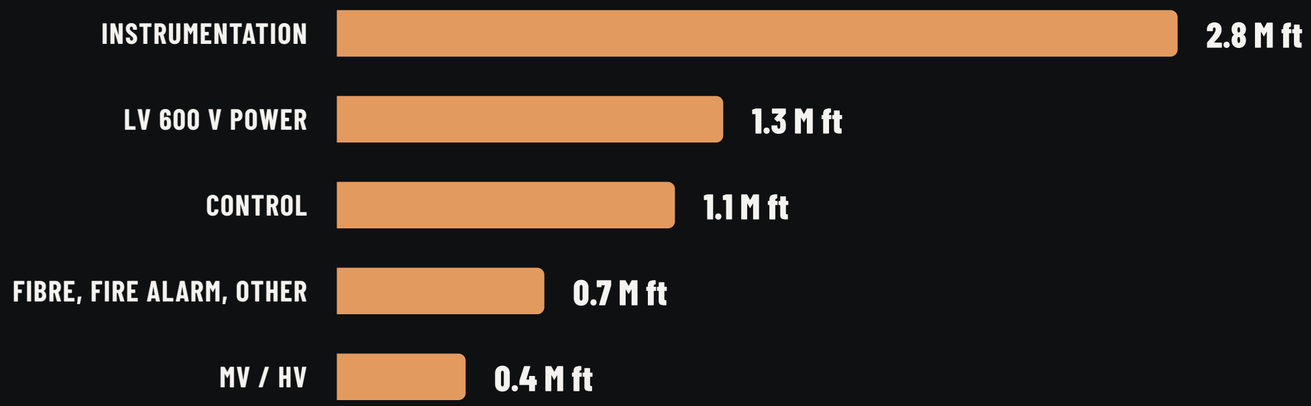
FEET OF CABLE · ABOUT 1,212 MILES

6,011
CABLE RUNS

21,217
CABLES

From the model's cable schedule: every circuit on the campus, from the 230 kV switchyard to the last thermocouple.

FOOTAGE BY CABLE TYPE



Most of the footage is small cable: instrument and control. MV is the high-value part, but it isn't the volume.

stephan.hardt@southwire.com · INSIDE POWER GENERATION 23/26

WHAT THE SPEC CALLS FOR

Every cable application on the HRSG, the requirements an owner's or EPC's cable specification writes for it, and the Southwire product built to meet them.

APPLICATION · WHERE	WHAT THE SPECIFICATION CALLS FOR	SOUTHWIRE
MV POWER, 15 KV FEED-PUMP MOTORS; THE T-R2A PRIMARY	1/C Cu, EPR insulation, **133 % level**, 90 °C, tape shield, PVC jacket. ICEA S-93-639, UL 1072 (MV-105). Terminations IEEE 48; acceptance test VLF (IEEE 400.2) / NETA ATS	**MV-105 15 kV, 220 mil NL-EPR 133 %, SIMpull · SPEC 46302**
480 V FEEDERS AND POWER T-R2A SECONDARY, LC-R2A TO MCCS, MOTORS AND MOVS	Cu XHHW-2, 90 °C wet / dry, sized at **125 %** of full load; voltage drop ≤ 3 %. ICEA S-95-658, UL 44 / UL 1277; 10+ parallel sets: compare with bus duct	**XHHW-2 1/C (SIMpull) · Type TC-ER 3/C + ground · SPEC 45252**
TRAY CABLE: POWER AND CONTROL EVERY TRAY BETWEEN THE E-HOUSE AND THE HRSG	Type TC-ER (UL 1277) so it may leave the tray as an exposed run to the equipment (NEC 336.10); sunlight resistant, direct burial; **IEEE 1202 / UL 1685** 70,000 Btu/h vertical-tray flame test	**Control TC-ER 14 AWG multiconductor · SPEC 45053**
HAZARDOUS AREA: GAS TRAIN, DUCT BURNER DUCT-BURNER FUEL SKID, BMS PANEL TO THE VALVES	Class I Div 2 wiring methods per NEC 501.10(B); MC-HL (UL 1569 + UL 2225) also qualifies for Div 1. Burner management per **NFPA 85**, duct burner UL 295, listed fittings	**ARMOR-X MC-HL, welded armor · SPEC 45352**
VARIABLE-FREQUENCY DRIVES SCR DILUTION-AIR BLOWERS, FD FANS	VFD-rated: symmetrical three-ground construction, overall shield or continuous armor for the common-mode current, XLPE insulation for reflected-wave peaks	**ARMOR-X MC-HL VFD · TC-ER VFD · SPEC 45352**
INSTRUMENTATION, 4-20 MA / HART DRUM LEVEL, PRESSURES, FLAME SCANNERS, ANALYZERS	**Individually shielded** twisted pairs + overall shield, drain wires; 300 V PLTC (UL 13, NEC 725) or ITC; ICEA S-73-532; segregated from power in tray	**Instrumentation PLTC, SPOS · SPEC 43303**
THERMOCOUPLE EXTENSION GAS-PATH, TUBE-METAL AND DUCT TEMPERATURES	Alloy matched to the thermocouple (type K = KX), ANSI MC96.1 / ASTM E230 colour code; **never copper**; shielded pairs; PLTC; 105 °C near the casing	**Thermocouple extension, PLTC multi-pair · SPEC 42345 / 42350**
FIRE ALARM, FIBRE, BUILDING WIRE F&G LOOPS, CEMS DATA, E-HOUSE LIGHTING	FPLR / FPLP (NEC 760, NFPA 72); single-mode fibre, armored in tray; THHN / THWN-2 in conduit (UL 83)	**SIMpull THHN / THWN-2 · fire-alarm and fibre cable**

CHECKED AGAINST A REAL PLANT

A recent US utility combined-cycle filing shows the same structure: **HRSG MCCs A / B** fed from 480 V switchgear through 2,500 / 3,300 kVA transformers off the auxiliary bus (6.9 kV there, 13.8 kV here).

DESIGN AND INSTALL

Raceway and cable design for generating stations: IEEE 422. Grounding: IEEE 80 / 665. Area classification: NEC 500, API RP 500. Field acceptance: NETA ATS.

stephan.hardt@southwire.com · INSIDE POWER GENERATION 24/26

PAIN POINTS AND THE ANSWER

TECHNICAL PAIN POINTS

- **MOSTLY SIGNAL CABLE** ~74 % of an HRSG's cables are instrument or thermocouple pairs; bad shield grounding or routing next to drives and MV makes noisy drum-level and flame-scanner signals that can trip the unit
- **THERMOCOUPLE EXTENSION ERRORS** ~84 runs per train: wrong alloy, reversed polarity or copper substituted; found as a bad tube-metal reading after startup
- **HEAT AND SUN** Hot casings, roof penthouses, drum platforms: 90-105 °C ratings, sunlight-resistant jackets, derating
- **MOVEMENT AND CYCLING** The HRSG grows inches when hot and cycles daily: slack at module joints, vibration, termination fatigue
- **GAS IN THE BOILER** Duct burners and fuel skids make Class I Div 2 areas; NFPA 85 burner management; conversions add it
- **BIG PARALLEL FEEDERS** 3,007 A transformer secondary = 40 cables of 500 kcmil; equal lengths in congested tray
- **DRIVES** SCR blowers and fans on VFDs: bearing currents and voltage spikes without VFD-rated cable
- **SCOPE GAPS** Vendor junction boxes, EPC home runs: legs fall between the two (our own schedule missed the 480 V secondaries)
- **SCHEDULE AND LABOUR** Cable is on the critical path, above all in an outage; electricians are scarce; no room to stage reels

VALUE PROPOSITION

- **ONE SUPPLIER, WHOLE SCOPE** MV-105, TC-ER, ARMOR-X MC-HL and VFD, instrumentation, thermocouple extension, THHN, fire alarm: one PO, one spec-compliance package (IEEE 1202, UL 1277 / 2225 / 13)
- **FASTER PULLS: SIMPULL** Pre-lubricated jacket cuts pulling force and time; parallel conductors on one reel go in one pull
- **STAGED FOR THE OUTAGE** Cut-to-length reels labelled per pull, SIMpull Cradles for one-person payoff: ready before the trip
- **ONE CABLE FOR HAZARDOUS AREAS** ARMOR-X MC-HL is rated Div 2 and Div 1: fewer conduits and seals, unaffected by reclassification
- **THE RIGHT CONSTRUCTION** VFD-rated cable for drives, individually shielded pairs, matched-alloy colour-coded thermocouple extension
- **ENGINEERING UP FRONT** Ampacity, voltage drop, spec review; takeoff ready: ~707 cables, ~157,000 ft per train, ~472,000 ft for three
- **SUSTAINABILITY** Reel returns and scrap recycling for cable cut back in brownfield outages

"On an HRSG, cable is three-quarters instrumentation and fully on the critical path. One spec-compliant supply, staged and fast to pull: wired on time, reading right the first time."

stephan.hardt@southwire.com

THREE THINGS TO REMEMBER

TAKE AWAY

1 **GAS IS GROWING AGAIN**

driven by data centres and the need for firm power.

2 **MOST OF A PLANT'S CABLE IS SMALL**

instrument and control; the big MV and power cable is the smaller share.

3 **SOUTHWIRE COVERS THE WHOLE PLANT**

one supplier, staged on SIMpull reels, spec-compliant from MV down to the thermocouple.

Southwire®
POWER GENERATION SOLUTIONS

Q&A

Questions from the chat. The recording and the 3D viewer link will be shared after the session.

Stephan Hardt
stephan.hardt@southwire.com